A Mixed Reality Virtual Clothes Try-On System

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***Abstract*—Virtual try-on of clothes has received much attention recently due to its commercial potential. It can be used for online shopping or intelligent recommendation to narrow down the selections to a few designs and sizes. In this paper, we present a mixed reality system for 3D virtual clothes try-on that enables a user to see herself wearing virtual clothes while looking at a mirror display, without taking off her actual clothes. The user can select various virtual clothes for trying-on. The system physically simulates the selected virtual clothes on the user's body in real-time and the user can see virtual clothes fitting on the her mirror image from various angles as she moves. The major contribution of this paper is that we automatically customize an invisible (or partially visible) avatar based on the user's body size and the skin color and use it for proper clothes fitting, alignment and clothes simulation in our virtual try-on system. We present three scenarios: i) virtual clothes on the avatar, ii) virtual clothes on the user's image and iii) virtual clothes on the avatar blended with the user's face image. We have conducted a user study to evaluate the effectiveness of these three solutions from the end user's perception of quality attributes, cognitive attributes and attitude towards using. The user study shows that among these three scenarios, the second one is most preferred by the users and for 50% of them the experience they had with our system was sufficient to make the purchase decision for the outfits they virtually tried-on.**

***Index Terms*—3D-2D alignment, body customization, mixed reality, skin color matching and real-time cloth simulation, virtual try-on.**

I. Introduction

PHYSICAL try-on of clothes is a time consuming procedure in retail shopping. It takes try-on of several clothes before the shopper can make decision on design, color and size of the apparel that satisfies her. Virtual try-on can help to speed-up the process as the shopper can see the clothes on her body without actually wearing them, or narrow down her selections before physical try-on. Furthermore, it can enhance the user's shopping experience through new features, such as side-by-side comparison of various clothes and simultaneous viewing of outfits from different angles. Interactive virtual try-on can also be an interesting feature of digital signage for advertisement and/or attracting crowds.

Some earlier systems using image processing techniques have been reported in literature. Hilsmann and Eisert described a dynamic texture overlay method from monocular images for real-time visualization of garments in a virtual mirror environment [1]. Subsequently, they proposed a method to segment the user's clothes and retexture them using extracted shading and shape deformation information [2]. The system presented in [3] took a user's picture in minimum clothes and covered it with those segmented from a picture of a model wearing target clothes. In [4], fiducial markers were used to change the texture of a user's shirt. In [5], a pre-generated 3D human model in target clothes was superimposed on a user's 2D picture. [6] presented a virtual clothing system, in which a user is scanned and registered to the system once, and then clothes can be simulated on the reconstructed model. The research in [7] enables a user to load his/her own 3D model, select 3D clothes from a catalogue and superimpose them on the model by interactive positioning. Frederic *et al.* presented a framework for a web-based solution using a generic database for dressing a user look-alike clothed avatar and simulating the clothes on the avatar [8]. Meng *et al.* and Wacker *et al.* proposed solutions for sewing of clothing patterns on a virtual 3D human character and viewing the simulation results of the clothes [9], [10]. Zou *et al.* implemented a web-based platform that allows interactive viewing of clothes simulation and selection of different hairstyles and other accessories on an avatar [11].

In recent years, some interactive virtual try-on solutions using augmented reality technique have been reported. A major challenging issue in AR based systems is the requirement of accurate pose recovery for fitting virtual clothes (or other accessories) on a user's image [4], [13]. Today with the release of a new generation of sensing technologies capable of providing high quality videos of both color and depth (e.g., Microsoft Kinect [14]), it provides an opportunity to dramatically increase the capabilities of virtual try-on solutions. [15] described an impressive virtual try-on system where a user's human model and cloth were reconstructed by deforming a SCAPE [12] model using a multiple-camera setup. Next, the skeleton obtained from a Kinect camera is mapped to the SCAPE model so the model can follow the user's movements captured by the Kinect camera. The advantage of this method is that the system can enable users to enjoy a private virtual try-on experience at their own homes. However, it is a time-consuming method and requires offline manual interventions to generate a 3D model of a user. Other impressive Kinect-based systems have been reported in [16]–[22]. Similar to [15], they also require the avatar model to be reconstructed manually offline.

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While a user's pose given by Kinect is sufficient for games and other applications, it is not accurate enough to align virtual clothes with the user's body properly. In this paper, we propose a robust alignment method to align the virtual clothes with the user's image properly using an automatically customized avatar that matches the user's body. Scaling of the virtual clothes to fit well on the user's body is another important requirement for a realistic virtual try-on. A global scaling does not work for users of different body types. We use a scaling method that scales different sections of the clothed avatar separately and also retains their connectivity by appropriately translating different parts after resizing. Since the clothes in our system are worn by an avatar, we can use certain body measurements of the user to automatically customize the avatar and the clothes to match to the user's body size. Besides inaccurate alignment and scaling, another major reason of lack of realism in current systems is that the actual clothes worn by the user remain visible from sides and the overlaid virtual clothes do not completely cover them. An example of this would be a virtual T-shirt augmented on a user's image wearing long sleeves dress. We propose a solution to solve this problem by using a partially invisible avatar. To understand the user's acceptability, we have also conducted a user-study to evaluate the effectiveness of our system from the end user's perspective and understand the user's acceptability.

Our contributions in this paper are summarized as follows:

1) We propose a system that allows the user for trying-on various virtual clothes with one consumable RGB-D sensor. As concluded from our user study, this system can help users to make their purchase decision without physically trying-on clothes.
2) We automatically customize an invisible (or partially visible) avatar and use it for proper clothes fitting, alignment and clothes simulation. This includes i) a body customization algorithm to modify the avatar based on the user's body sizes, ii) a method for adaptively matching the avatar's skin color to the user's face skin, and iii) a method for robustly aligning the customized avatar with the user's image.

This paper is organized as follows. Section II describes three virtual try-on scenarios of our system. Sections III–IV deal with the automatic avatar customization and skin-tone mapping algorithms, respectively. Section V presents a novel method for alignment of a 3D avatar with the user's 2D image. Section VI describes the implementation details with experimental results, followed by the user study results in Section VII. Related issues have been discussed in Section VIII. Finally, conclusion is given in Section IX.

II. The Proposed Try-On Scenarios

In this paper, we present three scenarios for virtual try-on of clothes, i.e., i) Avatar Only (AO): virtual clothes on an avatar, ii) Dress Only (DO): virtual clothes on a user's image, and iii) Hybrid Version (HV): virtual clothes on an avatar blended with a user's face image, as shown in Fig. 1(a)–(c), respectively. If a user is a female or male, the system will show a female or male avatar model, correspondingly. Fig. 2 shows the architectures of these three scenarios. A generic 3D avatar is first automatically customized based on the user's body sizes and its skin color is matched to the user's face skin color. We then use a novel algorithm to properly align the 3D customized avatar with the user's image in real-time. We use the simulation method in [23] for animating virtual clothes due to its speed and accuracy for physical garment simulation. In the first scenario, we remove the user's image from the screen and replace it with the clothed avatar. In the second version, 3D virtual clothes are augmented on the user's image without displaying the avatar. For the third version, we first segment out the user's image below the neck and replace it by a reconstructed background using a pre-captured background image. Then the customized headless avatar wearing the virtual clothes is rendered on the screen. As a result, the user sees her look-alike avatar on the screen wearing virtual clothes. In all three scenarios, the virtual clothes follow the user's movements, giving her a perception of trying-on virtual clothes in front of an actual mirror.

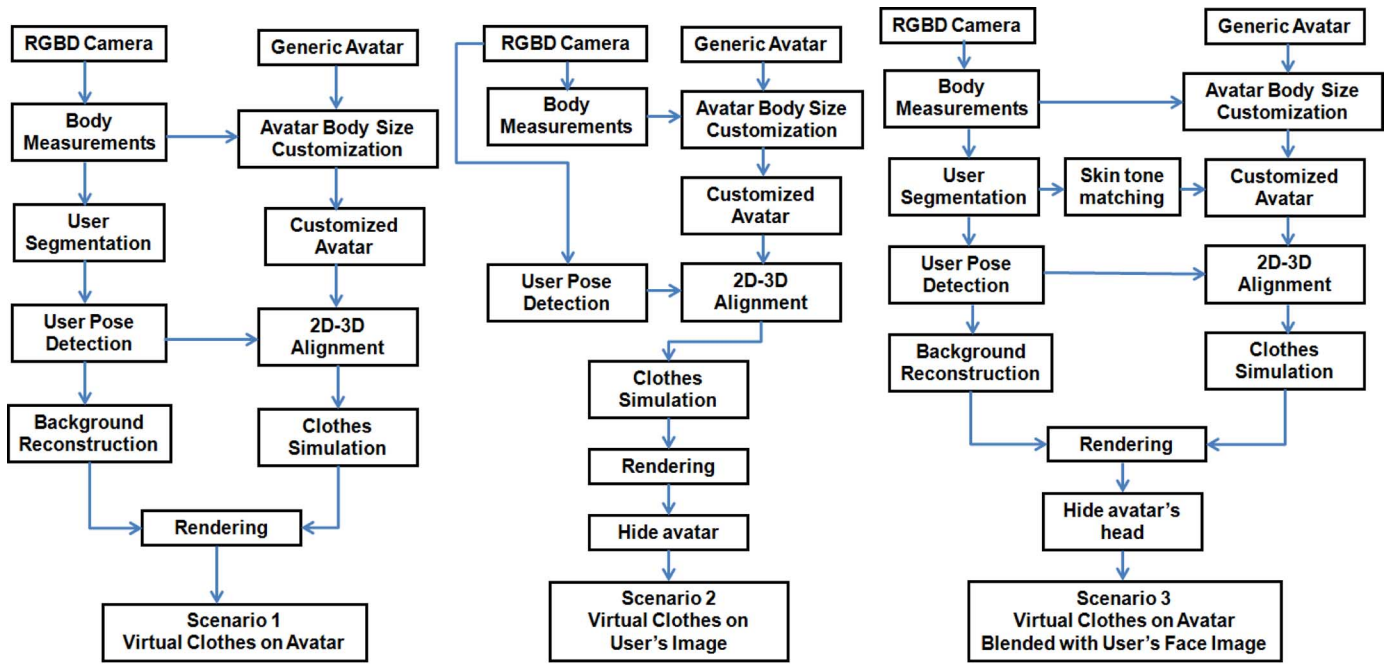
(a) (b) (c)

Fig. 1. The three proposed try-on scenarios: (a) Avatar Only (AO): virtual clothes on an avatar, (b) Dress Only (DO): virtual clothes on a user's image, and (c) Hybrid Version (HV): virtual clothes on an avatar blended with a user's face image.

III. Body Customization

Avatar customization can potentially be a strong tool in modeling and simulation applications. Rather than having to create models with different body shapes and sizes for a specific application, it is much economical in terms of time and effort, if a generic model can be modified realistically based on some parameters. We present a novel method to modify a generic avatar based on standard human body measurements for our virtual try-on system. Different from methods in [24], [25], which use free-form deformation and RBF-based deformation, respectively, we modify the avatar by scaling the vertices globally and locally. The method is faster than the existing methods, and is able to produce natural-looking results.

Based on our experimental results, an accurate user's look-alike avatar can be created based on twelve key human body measurements, i.e., height, shoulder width, bust girth, waist girth, hip girth, thigh girth, ankle girth, waist height, crotch height, knee height, upper arm length and forearm length. Some of these measurements (such as height, shoulder width, waist height and arm lengths) can be obtained directly from a RGB-D sensor, and can be used to customize an avatar which is suitable for virtual try-on. For a more accurate customization, we provide an option of keying-in



(a) (b) (c)

Fig. 2. The architectures of these three scenarios: (a) Avatar Only (AO), (b) Dress Only (DO), and (c) Hybrid Version (HV).

all the key measurements manually for each registered user. Also, these key measurements can be obtained using the recent Kinect-based human modeling methods [26]–[28].

The procedure of our automatic avatar body customization is summarized as follows:

1) Scale the model globally according to the user's height.
2) Scale the torso and the legs along the y-axis based on the user's waist, crotch, and knee heights. The scale s is computed as a function of y using piecewise cubic polynomial interpolation from the ratio of the user and the avatar heights, with $y = 0$ and $y = h_{bust}$ as the boundaries ($s = 1$), where $y = h_{bust}$ is the height of the avatar's bust.
3) Modify the torso and the legs based on the user's shoulder width, bust, waist, hip, thigh and ankle girths. At first the arms vertices are translated along the x-axis based on the user's shoulder width. Then the torso and the legs are scaled in the x-z plane using the model in Fig. 3, where s_g is the ratio of the user's girth to the avatar's one. The girth change for the leg is modeled as a uniform scaling in the x-z plane ($t = s_g$), while the torso girth change is modeled in the right figure of Fig. 3 ($t = 1$). s_g and t are computed as a piecewise cubic polynomial function of y, with the boundaries at $y = 0$ and $y = h_{neck}$, where $y = h_{neck}$ is the avatar's neck height. The scale k for the vertices (x, z) is computed as follows:

$$k = (s_g - t)\sin\theta + \sqrt{(s_g - t)^2 \sin^2\theta + 2s_g t - t^2} \qquad (1)$$

4) Modify the arms based on the user's upper arm and the forearm lengths.

It should be noted that the girth change for the torso is modeled as two internally tangent circles, even if the curve around the torso is approximately elliptical. The purpose is to simplify the modeling of the transition from the torso girth change to the leg girth change, which is modeled as two concentric circles. For validation of the proposed method, we used a number of 3D real human models taken from CAESAR database [29] as a reference, and customized our generic avatar to match their sizes. Fig. 4 shows some results (males and females) of our body customization method on a generic avatar for CAESAR database models. The resulting avatars looked realistic and matched the reference models well. Fig. 5 shows some experimental results of the clothed avatar based on the users' real sizes. It should be noted that the avatar modification is based on the position of the vertices. For example, for the torso modification, the algorithm selects and modifies all the vertices between $y = h_{knee}$ and $y = h_{neck}$, where y is the vertical axis. $y = h_{knee}$ and $y = h_{neck}$ are the heights of the knee and the neck, respectively. So the vertices of the clothes will be customized as well,

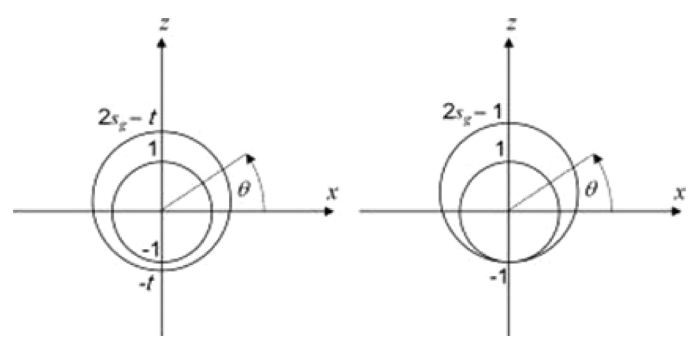


Fig. 3. The model for the girth change. The left figure shows the general model, where s_g is the ratio of the circumferences of the outer circle to the inner circle and t is the deviation from the uniform change ($t = s_g$ describes 2 concentric circles). The right figure is the model for the torso girth change ($t = 1$), where $(0, -1)$ is the location of the spine.

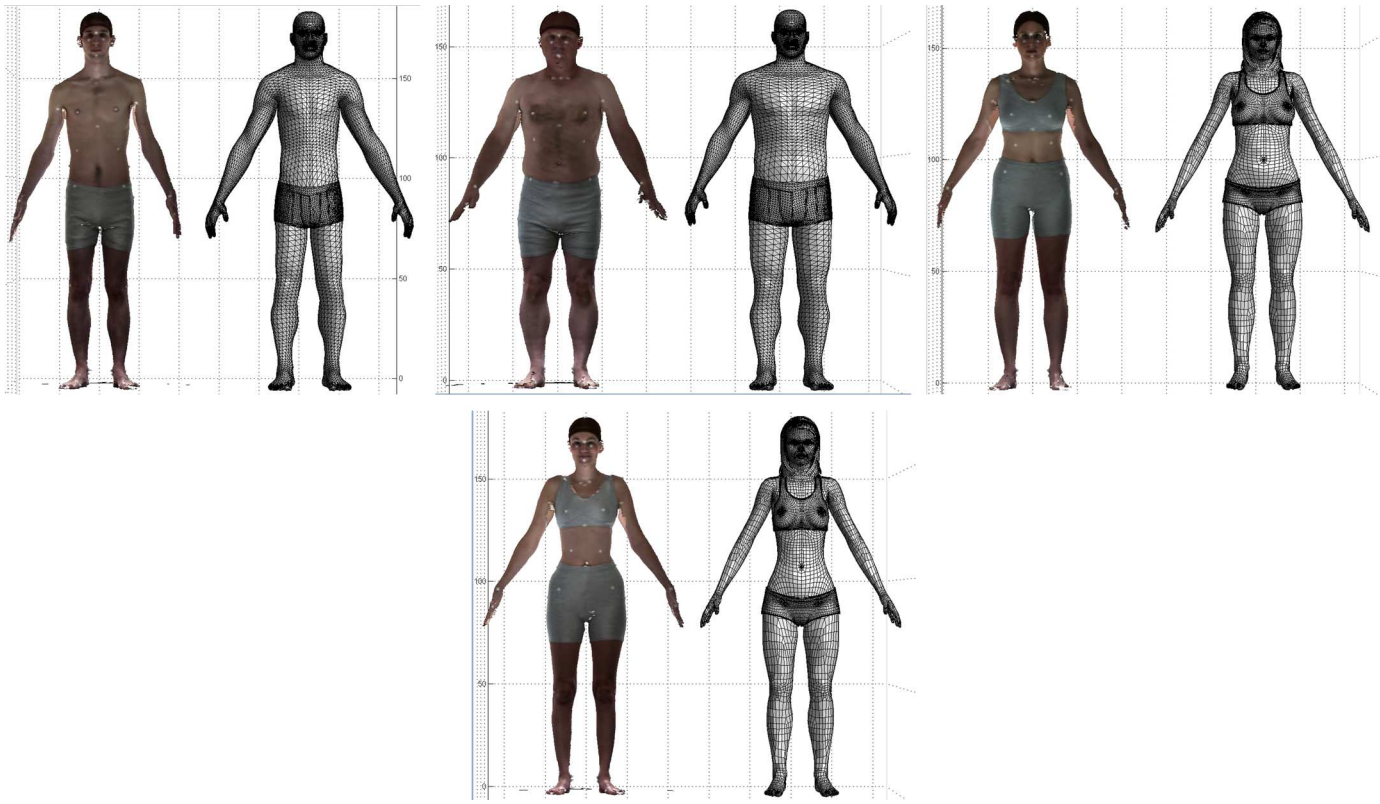


Fig. 4. Some automatic avatar body customization results using CAESAR database. The left in each figure is the real human model and the right is the corresponding customized avatar.

(a) (b) (c) (d)

Fig. 5. Some automatic avatar body customization results based on the users' real sizes: (a) (c) are the real user images, and (b) (d) are the corresponding customized avatars.

as the clothes have been fitted on the avatar during the offline design stage.

IV. Skin-Tone Matching

Besides editing the mesh geometry according to the user's sizes, skin-tone matching is another requirement for creation of a look-alike avatar. We use the user's actual face skin color to adaptively change the avatar's body skin color. We propose a method to transfer the user's skin tone to the avatar, which consists of three steps. In the first step, facial features are located using the active shape model (ASM) technique [30]. In the second step, we use piecewise linear curves to represent the cheek areas and extract cheek patches. Finally, we apply a global color transfer method [31] to shift the color of the face patches to the avatar body.

To acquire face skin patches, a common technique is applied to transfer the RGB image to the YCbCr color space and threshold the chrominance component. The pixels with their chromatic value falling in a certain range are classified as skin-color pixels. However, in most uncontrolled environments, some background clothes or hair may have similar colors as face. Moreover, the lips and the areas between eyebrows and upper eyelids (or even the eyebrows in some cases) are easily misclassified as skin areas which are not perfect skin patches to be used for skin-tone transfer. Another problem in skin tone extraction is that the appearance of skin is usually quite different under different viewing and lighting conditions. Additionally, due to different scattering properties of skin, cellular and 3D topographic structure of the face, there are usually highlights in the forehead, nose and chin areas.

We have observed that the cheek area is the largest flat skin area on the face, and is least affected by shadows of the nose, forehead and hair. It reflects the illuminated light uniformly compared to other areas on the face. Hence, we choose the cheek area as an ideal exemplar patch. We first detect the facial features using ASM, in which face shape can be formed with a set of landmarks that represent distinguishable points such as eye pupils, nose and lips etc. Total 76 landmarks are marked on the face, and joined by lines in a certain order, to get a piecewise linear representation of the face. We use 20 landmarks and their connecting lines to enclose the left and right cheeks and segment them from the face, as shown in Fig. 6(b) and (c).

To implement skin color transfer, we decompose the RGB color components of the face skin patches into lightness and

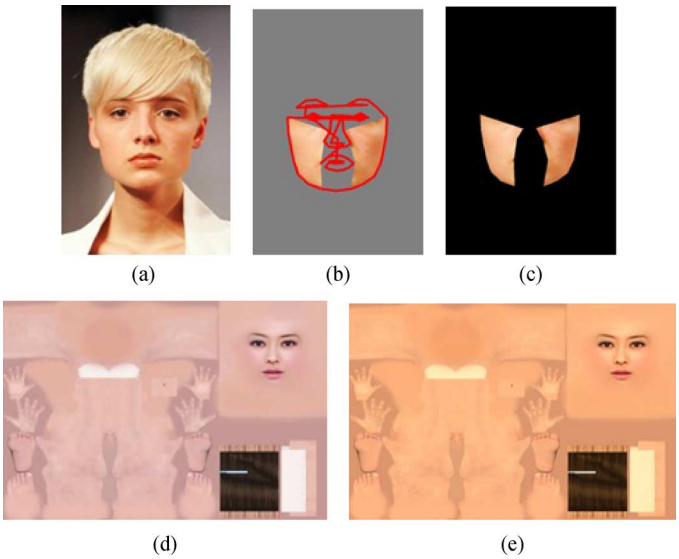


Fig. 6. (a) Original face image; (b) face cheek with landmarks and jointed lines; (c) "cut out" face cheek patches; (d) original avatar UV map and (e) new avatar UV map using cheek patches.

chrominance layers by converting them into the CIELab color space. The L channel is the lightness layer and the a, b channels are the color layers. We choose the CIELAB color space because it performs better than other color spaces in separating lightness from color, and is approximately perceptually uniform.

We compute the mean and the standard deviation of each channel for both the source and target images, i.e., the user's face skin patches and the avatar's UV map, and subtract means from the target image:

$$l_u^* = l_u - \overline{l_u}, a_u^* = a_u - \overline{a_u}, b_u^* = b_u - \overline{b_u} \tag{2}$$

where $\overline{l_u}$, $\overline{a_u}$ and $\overline{b_u}$ denote the corresponding mean values. The subscript u represents the avatar's UV map. We scale the data points of the synthetic image with weighting factors determined by the respective standard deviations and then add the average values of the user's face skin to generate new l, a, b color components, which are given:

$$l'_u = \frac{\sigma_f^l}{\sigma_u^l} l_u^* + \overline{l_f}, a'_u = \frac{\sigma_f^a}{\sigma_u^a} a_u^* + \overline{a_f}, b'_u = \frac{\sigma_f^b}{\sigma_u^b} b_u^* + \overline{b_f} \tag{3}$$

where σ_f and σ_u denote the standard deviations of the face skin and the UV map, respectively. The subscript f represents the face skin. Finally, we convert (l'_u, a'_u, b'_u) back to the RGB color space to give a new UV map which has a similar color tone as the face skin. A typical experimental result is shown in Fig. 6. We can see that the cheek patches are accurately extracted and represent the tone of the face well. The converted UV map of the avatar body has a similar skin tone to the user's face.

V. Aligning 3D Avatar With 2D User Images

In a virtual try-on system, accurate alignment between a 3D clothed avatar with a 2D user image stream is of crucial importance. In our system, the structure of the skeleton model obtained from an RGB-D camera and the built-in skeleton structure of the avatar are not the same. This is so that other RGB-D cameras with different skeleton models can also be used in our system. To align the 3D avatar with the user's body in every frame, we need to estimate a projection matrix that projects the avatar onto the user's image properly. The simplest way is to use the transformation matrix that maps one joint from the avatar's skeleton (e.g., the center of the shoulders) to its corresponding joint from the RGB-D camera skeleton. However, this solution is prone to misalignment errors for other body parts. In this section, we propose an efficient method which can properly align a 3D clothed avatar with a real user's image in real-time using an RGB-D sensor (e.g., a Kinect camera in our system) as described below.

A. Initial Alignment

To robustly align a 3D life-size avatar with a user's 2D image sequence, we need to establish a sufficient set of 3D-2D point correspondences between the 3D avatar model and the user's image in the current frame. Manual intervention is often required to obtain some initial 3D-2D correspondences [32]. However, automatic initialization is necessary in a virtual try-on system. In our system, we ask the user to stand and look straight in front of a display in a standard pose at the beginning for few seconds until the augmented clothed avatar is scaled properly and aligned with her image on the screen. We then save this image as a keyframe in our system that will be used later for continuous body tracking and robust alignment.

Let's assume the user's image point and its corresponding 3D point from the avatar are denoted by $\mathbf{m} = (u, v)^T$ and $\mathbf{M} = (X, Y, Z)^T$, respectively. We use homogeneous vectors to represent a 2D image point $\mathbf{m}$ and a 3D point $\mathbf{M}$ as follows:

$$\mathbf{m} = (u, v, 1)^T,\ \mathbf{M} = (X, Y, Z, 1)^T \tag{4}$$

We would like to find a projection matrix $\mathbf{P}$ that satisfies the following equation for the 3D point $\mathbf{M}$ and the 2D user's image point $\mathbf{m}$:

$$\rho\mathbf{m} = \mathbf{K}[\mathbf{R}|\mathbf{t}]\mathbf{M} \tag{5}$$

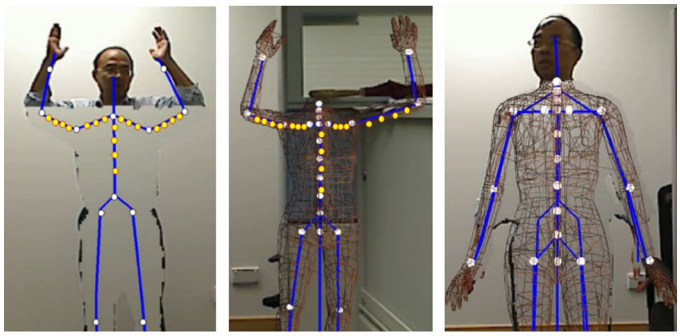
where ρ is an arbitrary factor, $\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]$ is the projection matrix and $\mathbf{K}$ is the camera intrinsic matrix. For the initial 3D-2D alignment, the purpose is to estimate the parameters $\mathbf{P}^j$ for the current frame j that minimizes the sum of squared distances between the projections of the 3D points on the avatar and the corresponding image points. We can estimate $\mathbf{P}^j$ using the following equation:

$$\mathbf{P}^j := \arg\min_{\mathbf{P}^j} \{w_J E_J + w_B E_B + \lambda \|\mathbf{P}^j\|^2\} \tag{6}$$

where the first two terms are defined below:

$$E_J = \left\{ \sum_{\mathbf{M}_i \in J_{Avatar}, \mathbf{m}_i \in J_{User}} \|\mathbf{m}_i - \zeta(\mathbf{P}^j, \mathbf{M}_i)\|^2 \right\} \tag{7}$$

$$E_B = \left\{ \sum_{\mathbf{M}_i \in B_{Avatar}, \mathbf{m}_i \in B_{User}} \|\mathbf{m}_i - \zeta(\mathbf{P}^j, \mathbf{M}_i)\|^2 \right\} \tag{8}$$



(a) (b) (c)

Fig. 7. (a) and (b) Joints and bone sampling points on the user's skeleton and the avatar's skeleton, and (c) an initial alignment result. The joints and sampling points are denoted by the white and yellow dots, respectively.

$\zeta(\mathbf{P}^j, \mathbf{M}_i)$ denotes the projections of 3D avatar points $\mathbf{M}_i$ given the parameters $\mathbf{P}^j$ in the jth incoming frame. J_{Avatar} and B_{Avatar} are the sets of joints and sampling points on bones from the avatar's skeleton respectively. Similarly, J_{User} and B_{User} are the sets of the projected RGB-D skeleton joints and sampling points on bones on the 2D input image.

It should be noted that we currently only use the sampling points on the upper arms, shoulders and spine as the RGB-D camera provides more accurate pose estimation on upper body parts. $\lambda\|\mathbf{P}^j\|^2$ is a regularizer which is used to avoid over-fitting and λ is set as 0.0002. w_J and w_B are the weights for the first two terms. Equation (6) can be solved using the Levenberg-Marquardt (LM) minimization method [33]. Fig. 7(c) shows an example of the initial alignment result.

B. Robust 3D-2D Alignment

To improve robustness and smooth the movements of the rendered clothes, it is desirable to have more 3D-2D point correspondences $\{\mathbf{M}_i, \mathbf{m}_i^j\}$ for the current frame j. In our system, we rely on silhouette contours and feature points to establish more 3D-2D correspondences.

Contour correspondences are established between the projected surface of the avatar and the user's image silhouette by searching for the 3D vertices of the avatar whose projections are the nearest to the user's contour points. We use the RGB-D camera to segment the user and extract the user silhouette. We then use the initial alignment results, as mentioned in Section V(A), to obtain contour-based 3D-2D point correspondences $\{\mathbf{M}_i, \mathbf{m}_i^j\}$. To meet real-time performance, we only consider the contours of the user's two shoulders, as show in Fig. 8. For the feature point correspondences, we need to establish the 2D-2D correspondences in real-time between the keyframe and the current frame j. We use the keypoint recognition method [34], where matching is formulated as an image classification problem. In this method, each class is trained with all the possible appearances that an image feature may take under large perspective and scale variations. The major advantage of the learning-based matching method is that it is very fast and is therefore suitable for applications that require real-time performance. Fig. 9 shows some 2D-2D correspondences results using the method in [34]. It should be noted that for aligning the customized avatar with the user's image, we only consider the feature points below the user's head, as we do not customize the avatar's head based on the user's head image.

Fig. 8. Corresponding contour points between the avatar's and the user's shoulders denoted by white and red crosses, respectively. For the purpose of visualization, the clothed avatar was not shown in this figure.

Fig. 9. Results of the learning-based real-time tracking method [34]. The left in each pair is the keyframe and the right is the incoming frame.

In both cases, the 2D correspondences are associated with a 3D vertex $\mathbf{M}_i$ on the avatar, thus yielding the 3D-2D correspondences $\{\mathbf{M}_i, \mathbf{m}_i^j\}$. We then extend (6) to estimate $\mathbf{P}^j$ as follows:

$$\mathbf{P}^j := \arg\min_{\mathbf{P}^j}\{w_J E_J + w_B E_B + w_C E_C + w_F E_F + \lambda\|\mathbf{P}^j\|^2\} \tag{9}$$

The definitions of E_C and E_F are the same as E_J and E_B, which define the errors based on contour and feature point correspondences, respectively. We estimate $\mathbf{P}^j$ in (9) using the LM method and the RANSAC technique. The weights w_J, w_B, w_C and w_F are set as 0.4, 0.4, 0.1 and 0.1, respectively, based on our experiments.

Fig. 10 shows two example results of our proposed method in automatically aligning the 3D avatar with the user's poses. To better illustrate the alignment performance, we only show the avatar's mesh on the user's image in the figure. As can be seen, the avatar's mesh, which has been automatically customized, follows the user's pose, and has been well aligned to the user's image in the incoming frame.

The proposed method is based on the information from the current frame. Hence, there will be no accumulated errors over time. It works well as long as the RGB-D camera is able to detect the user's poses. The tracking will be lost when the camera

Fig. 10. Examples of 3D-2D alignment results using the proposed method.

is unable to detect the user's poses properly, such as large rotations. It can be recovered from failed tracking once the RGB-D camera detects the user's poses again. It should be noted that due to the synchronization issue of the Kinect camera, there is a slight delay between the RGB stream and the depth stream. Hence, when the user performs a fast movement, there will be a noticeable lag in the movement of the clothed avatar.

In order to measure the accuracy of our alignment algorithm, we conducted the following experiment. We asked 10 female users to stand in front of our system and 10 virtual garments were overlaid one by one onto their images shown on the screen (the Dress Only scenario). Next, the users were asked to manually move each virtual garment up/down and left/right until it was best aligned to their liking in a standing pose. We then compared the manual alignment data for these 10 users and 10 garments with the results of our automatic alignment method. Table I shows the mean average error (MAE) between the manual alignment and our automatic method for each garment as well as the standard deviations (STD) on both X and Y axes. All numbers in this table are in meters. As can be seen in Table I, the average error of our alignment method in X direction is 5 mm while the alignment error is much less in Y direction (only 0.3 mm). The reason for higher alignment accuracy in Y-axis as compared to the X-axis, is because we use the shoulder points in our automatic alignment method and all these points are more or less distributed horizontally, while other feature points affect both X and Y directions equally. Hence, the algorithm can align the virtual clothes more accurately in Y direction than in X direction.

TABLE I
MEAN AVERAGE ERRORS AND STANDARD DEVIATIONS BETWEEN THE MANUAL AND OUR AUTOMATIC ALIGNMENT METHODS (IN MILLIMETER)

Virtual Garment	MAE_X	MAE_Y	STD_X	STD_Y
1	6.3	-7.0	2.5	1.47
2	7.2	7.2	6.6	0.58
3	8.2	-3.0	6.4	0.41
4	7.4	9.2	5.0	1.22
5	3.6	0.4	5.8	0.60
6	7.4	2.4	8.8	0.40
7	6.2	-2.0	3.4	0.83
8	-0.4	4.6	5.5	0.57
9	3.8	-1.6	2.3	0.45
10	2.2	-7.2	2.7	0.74
Average	5.2	0.30	4.9	0.73

Fig. 11. Some virtual trying-on results. The first, second and third rows show the results of the three versions, i.e., AO, DO and HV, respectively.

VI. IMPLEMENTATION

The virtual try-on system presented in this paper has been implemented under Visual Studio 2010 on a 2.53GHz Intel Xeon(R) with 24 GB RAM and runs in a single thread. We use a Kinect camera for pose detection, body measurements, user segmentation and face skin color detection.

At the beginning, the user stands in front of a display in a standard pose. The system automatically establishes relevant 3D-2D correspondences based on a keyframe. In addition, the user's body size (i.e., height, shoulder width, waist height and arm lengths) and the user's face skin color are extracted using the Kinect camera. The avatar is then automatically customized according to the extracted sizes and the skin color. As mentioned in Section III, the user can also key-in more body sizes for a more accurate avatar customization. Next, the user can select her favorite virtual clothes for virtual try-on. The selected virtual clothes will be aligned on the user's image, physically simulated, and rendered in real-time.

Fig. 11 shows some example results of our system for the three scenarios. In the first row, we show a customized clothed avatar. The second row shows the virtual clothes on the actual user's image. In the third row, the avatar's skin color automatically matches to the user's face skin color using the method described in Section IV. In this example, the clothes were fitted on the avatar during the offline design stage. The user can see the virtual trying-on results with various clothes from different angles based on here movements.

Computation time is often critical to many real-time applications. The average computation time for each frame is about 110 milliseconds. Table II shows the total computation time for

TABLE II
THE RUNNING TIME PERFORMANCE (MILLISECOND)

Background Reconstruction	3D-2D Alignment	Rendering
47	0.6	63

three time-consuming stages, i.e., user segmentation and background reconstruction, 3D-2D alignment, and rendering. Thus, the average frame rate for the system is about 9.1 fps. The computation time for the 3D-2D alignment is less than 1 millisecond. The most time consuming stages are the rendering and background reconstruction. The rendering time largely depends on the complexity of the clothes and the avatar. The background reconstruction algorithm utilizes the user detection result of the RGB-D camera and replaces the detected user image by a pre-captured background image. The main computation time is occupied by a pixel-by-pixel replacement procedure. The computation time and the inpainting artifacts of the background reconstruction results might be reduced using related efficient image replacement methods, such as the method in [36].

VII. USER STUDY

In this section we describe our user study to evaluate the effectiveness of three virtual try-on solutions from end user's perception of quality attributes (QA), cognitive attributes (CA) and attitude towards using (ATU).

A. User Study Design

Conducting user studies to understand perceptions of usability and usefulness of new technologies early allow rectifications to be done to avoid incurring high costs of remedial works in the developmental stage [35]. A within-subjects user study was conducted to determine the advantages and disadvantages of each version. We proposed an inventory of fifteen attributes covering perceptions of QA, CA and ATU as a starting point for measuring and comparing the perceptual and usability issues between the three versions [37]–[40]. QA covers factors relating to areas of concern that have the application potential to assist a purchase decision, particularly in the reliability, accuracy and user centric issues. CA relates to the attributes concerning the mental processes involved when using the system, including attention, learning, decision making and emotive elements. ATU relates to the set of attributes resulting from a presence of perceived ease of use and perceived usefulness of the system. These fifteen statements were represented in randomly ordered questionnaire statements, as shown in Table III, using a 5-point Likert scale (i.e., 1 = Strongly disagree; 5 = Strongly agree), to describe the degree that participants agree or disagree with the statements after interacting with each version of the system. Items in the questionnaire were designed with a mixture of positive and negative worded statements to minimize biasness due to response acquiescence.

In the user study, participants were presented with the same scenario for each of the three versions, which stipulate them to use the system to try-on five 3D virtual clothes. A total of 14 participants with an average age of 31.21 years ($SD = 10.61$), took part in the study. The occupations of the participants were all in the engineering domain and none were fashion-related. Only 1 participant reported using a virtual reality try-on system before (virtual makeup system), with 71% reported having shopped online for apparels. When asked if it is a must to physically try-on the clothes before making a purchase, 36% responded yes, 14% responded no, while 50% commented that this depends on a number of factors, such as the cost of the item and availability of a fitting room. A measure of reliability using Cronbach's Alpha was conducted on the questionnaire and results showed that all 15 attributes were highly reliable with an alpha ranging from 0.82 to 0.93. Spearman Rho correlation analysis of the dimensions to the overall means of each version was also conducted. Results showed that all three dimensions (QA, CA, and ATU) were significantly positively correlated to their respective overall mean scores, with the dimension ATU showing the strongest correlation for both the AO ($r = .944$) and HV ($r = .966$) versions, and CA showing the strongest correlation for the DO ($r = .919$).

TABLE III
QUESTIONNAIRE IN OUR USER STUDY

Quality Attributes (QA)	1. I found it user-friendly 2. I felt that using it was reliable in assisting me to make a purchase 3. I felt that this needs a lot of improvement 4. I liked its appearance 5. It did not match my expectations 6. I felt that using it was accurate in assisting me to make a purchase
Cognitive Attributes (CA)	7. I found it confusing to use 8. I got flustered when using it 9. I felt in control throughout when using it 10. I felt under stress while using it
Attitude Towards Using (ATU)	11. I will be happy to use it again 12. I enjoyed using it 13. I will recommend others to use it 14. I will frequently use it in future if I have the chance 15. I will say positive things about it to future users

B. Questionnaire Results

Fig. 12 shows results of the overall means and the corresponding constructs for the three versions. Overall, the users responded positively to the DO and HV versions, with the AO version scoring the lowest. Cognitively, all three versions were perceived positively at above means of 3.0, while qualitatively, all three versions were perceived a lower means of less than 3.0. Participants' ATU was perceived to be highest for the DO version and lowest for the AO version. From our user study, no significant difference was found in the dimensions of QA

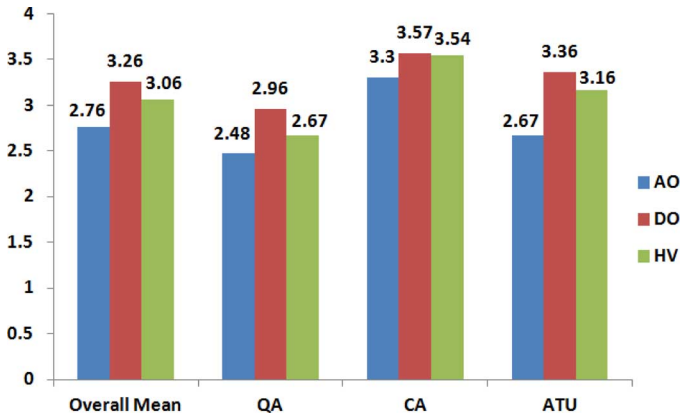


Fig. 12. Graphical illustration of overall and dimensional mean scores per version.

TABLE IV
RESULTS OF RANKING OF VERSIONS

Rank	Avatar Only	Dress Only	Hybrid Version
1st place	2	10	2
2nd place	3	3	9
3rd place	10	2	3

and CA across all versions using the Wilcoxon test [41] and the non-parametric Friedman test [42].

To assess if the use of such a system would translate to a purchase intention, participants were asked at the end of each version if they would purchase the outfits based on the experience they just had. For the AO version, only one participant said Yes, while the majority 13 said No. Four said they would purchase the outfits based on the experience that they had with the HV version, as it allows them to "see their face" and gives them allowance to assess "if the dress suits their skin tone or hairstyle". Interestingly, among all the participants trying the system, 50% of them said the DO version would suffice to exert a purchase intention, by justifying that "I can see most of my body parts (legs, arms etc.)", "at least it seems more realistic virtually.", "This definitely beats online shopping experience as I am able to see for myself how the dresses really look like on my body, and not by imagination." This higher rating of the DO version in terms of affecting purchase decision also resonated with the user's perception of system reliability in making a purchase. The average score of the DO version for the second quality attribute was 3.14, whereas the HV and AO versions scored 2.43 and 2.07, respectively.

C. Debrief Interview

At the end of all versions experienced in the user study, an interview was conducted to gather more information on the participants' perceptions and experiences of the versions.

As shown in Table IV, the majority of participants ranked DO versions as the preferred amongst the three. A majority of participants (71%) ranked the AO version as the least favorite. It should be noted that the total number does not equal to 14 (number of participants) due to similar ranking accrued for two versions.

TABLE V
DESCRIPTIVE ANALYSIS OF THE VISUAL QUALITY EVALUATION

	Means	Standard Deviations
AO	2.78	0.642
DO	3.42	1.186
HV	3.07	0.994

Two participants liked the AO version, commenting that the image gives them an overall image of how the outfit would look like. Some of the dislikes about the AO version were that they could not gauge if the outfit would look good on themselves. 10 of the 14 participants chose the DO version as their favorite, citing reasons that this version is next best to trying on the outfits physically. The fact that users only see the image of the dress being superimposed onto themselves on the screen gives them a better perception of how their skin tone would match to the color of the outfit. The dislikes for the DO version were mainly because users can still see the dress that they were wearing underneath the superimposed image. Hence, it could not give a realistic and accurate impression of the user wearing the outfit. Two participants ranked the HV as their favorite, as they could better assess themselves wearing the outfit, and at the same time, check if the outfit matches their hairstyle and faces. Some participants felt that the hybrid version did not give an accurate overall image of the outfit on them as the body was that of an avatar. Hence, the skin tone of the avatar did not perfectly match to that of the user and as a result, could not properly give them a good assessment of themselves wearing the dress. In summary, it would seem that the DO version has performed better than the other two versions, with the participants having closely similar cognitive perceptions for both the DO and HV versions.

We evaluated the visual quality from the end user's perception of visual quality attribute (i.e., "I liked its appearance." in QA). Table V shows the descriptive analysis of the visual quality for the three versions. Cognitively, the HV and DO versions were received positively by users (their means were above the average of 3.0) while the AO version was scored by 2.78, lower than average. The respondents expressed various views in terms of the visual quality between the versions. For the AO version, respondents thought that it only gives an overall image of the avatar wearing the dress and following human movements. Some respondents thought the overall visual quality in the DO version is "more flattering" and "more realistic", as respondents can see how exactly their skin tone match the dress. It gives a more accurate interpretation of the dress. For the HV version, some respondents felt "a complete image of the outfit" and they can also see their faces. However, some felt like that sometimes they were "fitting into someone else's body.

VIII. DISCUSSION

In our system, we implemented a series of techniques and combined them for virtual try-on application. The user study showed that among the three proposed solutions (DO, AO, HV) the DO version was most preferred by the users. Although the HV solution provides more realism than others, it is currently perhaps at the stage that falls into the uncanny valley. There

are several factors that currently affect the performance of the proposed system:

1) Large body rotation: For virtual or actual clothes try-on, users usually like to rotate and see the try-on results from different angles. However, large rotation cannot be detected reliably with only one RGB-D camera. As a future work, we plan to use multiple RGB-D cameras to detect user's large rotation more accurately.
2) Body customization: Currently, we use a Kinect camera as our RGB-D sensor to obtain some measurements from the user's body for our automatic avatar customization. However, in order to match the avatar shape with the user's body shape, more data is required. Recently, Weiss *et al.* proposed a method which is able to accurately model a human from an RGB-D camera [26]. This method is time-consuming and requires manual initializations. It remains an open problem on developing solution for fast, automatic and accurate modeling of a 3D human body .
3) Skin tone mapping: We currently use a one-time procedure to change the avatar's skin color during the initialization stage under the assumption of constant lighting conditions. However, in a real environment, the skin color of the customized avatar still has yet to match the user's skin color perfectly. From our user study, the participants raised the difference between the avatar's skin tone and their skin color. For the HV scenario, some users felt that this color mismatch negatively affects the feeling of trying the virtual clothes on our system. Especially this issue is more problematic in the neck region when the user's head image linked with the avatar's body. However, to reduce the artifact, currently we use alpha blending technique to have a smooth color transition in the neck region. Nevertheless, it remains an open problem on how to seamlessly connect the user's face to the neck of the avatar without any noticeable mismatch in the color and size in the junction line.

IX. Conclusion

In this paper, we described a mixed reality based virtual clothes try-on system using one consumable RGB-D camera (i.e., Kinect). We proposed a series of novel techniques and integrated them into a virtual try-on system, which includes three scenarios, i.e., (1) virtual clothes on the avatar, (2) virtual clothes on the actual user's image, and (3) virtual clothes on the avatar blended with the user's face image. The major contribution of this paper was that we automatically customized an invisible (or partially visible) avatar based on the user's body size and used it for clothes fitting, alignment and clothes simulation in our virtual try-on system. A user study was also conducted to evaluate the effectiveness of the system and the results showed that it can be effective in enabling customers' purchase decision.

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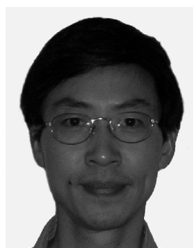

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